Heart Disease Prediction Using Machine Learning

1) Project Overview

This project focuses on predicting the likelihood of heart disease using Machine Learning based on patient clinical records. The complete workflow covers data collection, preprocessing, exploratory data analysis (EDA), feature engineering, model building, model comparison, application development, and cloud deployment.

The final solution was deployed as an interactive Streamlit web application on Render, where users can input health-related parameters and receive instant predictions.

This project demonstrates strong practical skills in:

- Data cleaning
- Feature engineering
- EDA and visualization
- Classification modelling
- Model evaluation
- Pickle serialization
- Streamlit app development
- Deployment on cloud

2) Problem Statement

Heart disease is one of the leading causes of death worldwide. Early prediction using patient health data can help in faster diagnosis and preventive treatment.

The goal of this project is to build a **binary classification system** that predicts whether a patient is likely to have heart disease based on medical attributes such as:

- Age
- Sex
- Chest pain type
- Blood pressure
- Cholesterol
- ECG results
- Exercise-induced angina
- ST slope
- Other clinical indicators

3) Objective

The main objectives of this project are:

1. Build an accurate heart disease prediction model.
2. Compare multiple machine learning algorithms.
3. Identify the best-performing model.
4. Deploy the final model as a user-friendly web app.
5. Present the model performance visually for end users.

4) Dataset Information

The dataset was collected from **Kaggle** and loaded into **Jupyter Notebook** for analysis.The Size Of Data set is 918 rows*12 Columns.

Features used

- Age
- Sex
- ChestPainType
- RestingBP
- Cholesterol
- FastingBS
- RestingECG
- MaxHR
- ExerciseAngina
- Oldpeak
- ST_Slope
- HeartDisease (Target)

	Age	Sex	ChestPainType	RestingBP	Cholesterol	FastingBS	RestingECG	MaxHR	ExerciseAngina	Oldpeak	ST_Slope	HeartDisease
0	40	M	ATA	140	289	0	Normal	172	N	0.0	Up	0
1	49	F	NAP	160	180	0	Normal	156	N	1.0	Flat	1
2	37	M	ATA	130	283	0	ST	98	N	0.0	Up	0
3	48	F	ASY	138	214	0	Normal	108	Y	1.5	Flat	1
4	54	M	NAP	150	195	0	Normal	122	N	0.0	Up	0
...	...	...	...	...	...	...	...	...	...	...	...	...
913	45	M	TA	110	264	0	Normal	132	N	1.2	Flat	1
914	68	M	ASY	144	193	1	Normal	141	N	3.4	Flat	1
915	57	M	ASY	130	131	0	Normal	115	Y	1.2	Flat	1
916	57	F	ATA	130	236	0	LVH	174	N	0.0	Flat	1
917	38	M	NAP	138	175	0	Normal	173	N	0.0	Up	0

918 rows × 12 columns

Target Variable

- **HeartDisease**
    - 0 = No heart disease
    - 1 = Heart disease

5) Tools and Technologies

- Python
- Pandas
- NumPy
- Matplotlib / Seaborn
- Scikit-learn
- Jupyter Notebook
- VS Code
- Streamlit
- Pickle
- Render

6) Data Preprocessing

Data preprocessing was one of the most important steps in this project.

6.1 Data Loading

The dataset was imported from Kaggle and loaded into Jupyter Notebook using Pandas.

6.2 Data Inspection

Initial checks included:

- df.info() to inspect data types and null values
- df.describe() to understand statistical summary

During the descriptive analysis, it was found that:

- **RestingBP minimum = 0**
- **Cholesterol minimum = 0**

These values are medically impossible and indicate invalid entries.

6.3 Handling Invalid Values

Since BP and cholesterol cannot realistically be 0:

Cholesterol Treatment

Cholesterol values were replaced using the **median grouped by Sex**, because cholesterol levels often vary across gender.

Logic:

- Group by Sex

- Replace 0 values with sex-wise median cholesterol

Resting Blood Pressure Treatment

For BP, 0 values were replaced using the **mean blood pressure** of the dataset.

This improved data quality and prevented the model from learning unrealistic patterns.

6.4 Null and Duplicate Check

The dataset was checked for:

- Missing values
- Duplicate rows

Result:

- **No null values found**
- **No duplicate values found**

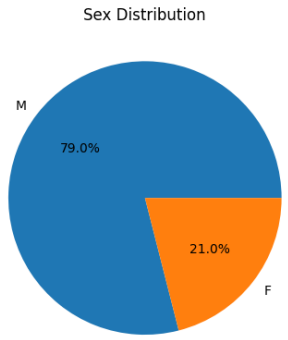
7) Exploratory Data Analysis (EDA)

EDA was performed to understand relationships between patient features and heart disease.

Note: Add your charts/screenshots manually in the places below.

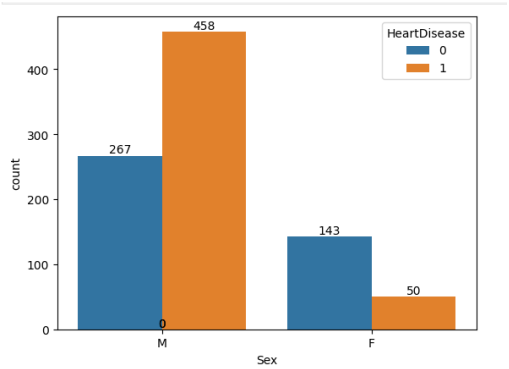
7.1 Sex Distribution

A pie chart was created to visualize male vs female distribution.



7.2 Sex vs Heart Disease

A bar chart was used to compare heart disease cases across gender.



7.3 Age Group Analysis

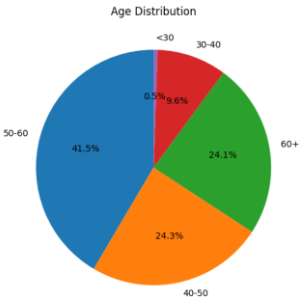
Age groups were created in a new data frame (df1) to better analyze risk patterns.

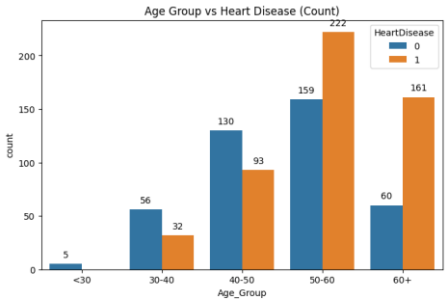
Examples:

- Young
- Middle-aged
- Senior

Then visualized:

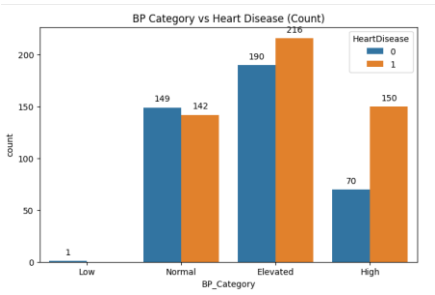
- Age group distribution
- Age group vs heart disease





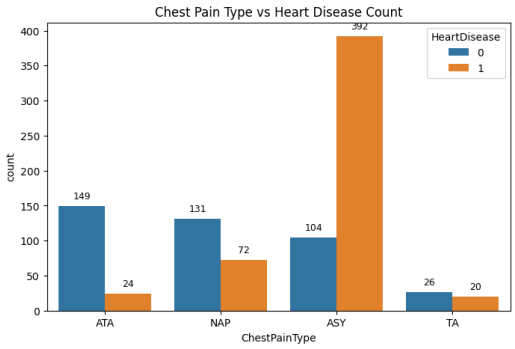
7.4 Blood Pressure Group Analysis

Blood pressure was grouped into categories and visualized against heart disease.



7.5 Chest Pain Type vs Heart Disease

Chest pain categories were analysed to understand their relationship with disease occurrence.



8) Feature Encoding

Categorical variables were converted into numerical values.

Encoding Mappings

- **Sex**: M = 0, F = 1
- **ChestPainType**:
  - ATA = 0
  - NAP = 1
  - ASY = 2
  - TA = 3
- **RestingECG**:
  - Normal = 0
  - ST = 1
  - LVH = 2
- **ExerciseAngina**:
  - N = 0
  - Y = 1
- **ST_Slope**:
  - Up = 0
  - Flat = 1
  - Down = 2

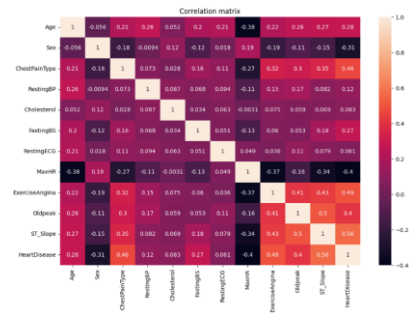
After encoding, all data types were converted into **integer format**.

9) Correlation Analysis

A correlation heatmap was created to identify the strength of relationships between features and the target variable.

This helped in:

- Understanding feature importance
- Detecting highly related variables
- Supporting model selection



10) Model Building

Four machine learning classification models were trained and evaluated.

Models Used

1. Logistic Regression
2. Support Vector Machine (SVM)
3. Decision Tree
4. Random Forest

Train-Test Split

The dataset was split into:

- Training set
- Testing set

for fair evaluation.

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11) Model Performance

The following accuracy scores were achieved:

Model	Accuracy
Logistic Regression	0.8586
SVM	0.8316
Decision Tree	0.8097
Random Forest	**0.8641**

Best Model

The **Random Forest Classifier** delivered the highest accuracy of **86.41%**, making it the best-performing model for deployment.

12) Model Serialization

All trained models were saved as **Pickle (.pkl) files**.

This allowed the trained models to be reused inside the Streamlit application without retraining.

Saved artifacts include:

- Logistic Regression pickle
- SVM pickle

- Decision Tree pickle
- Random Forest pickle

A user-friendly **Heart Disease Predictor App** was built using **Streamlit** in VS Code.

App Features

The application contains **two tabs**:

13.1 Prediction Tab

Users can enter patient values for all input variables.

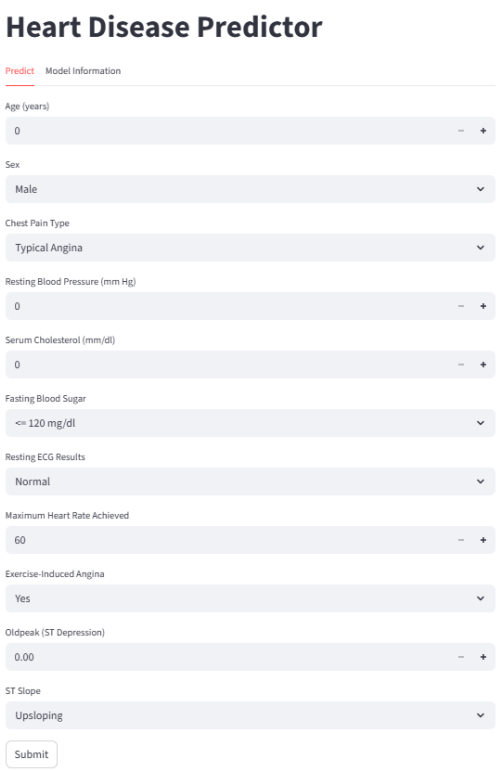
The model then predicts:

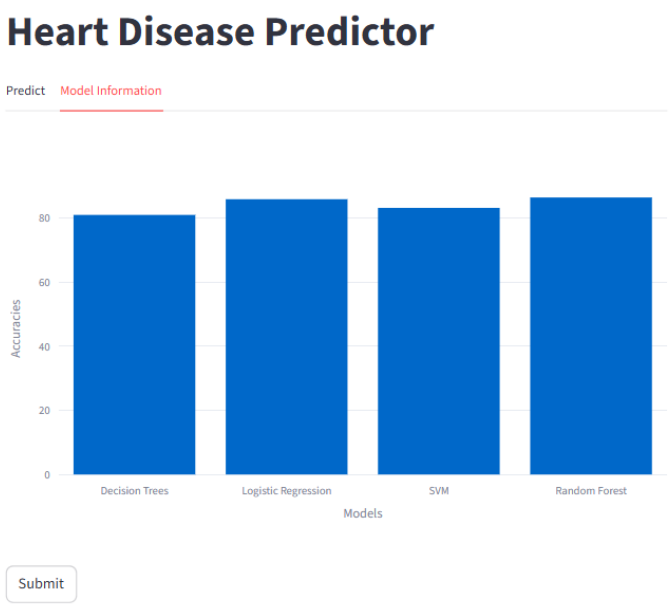
- Presence of heart disease
- No heart disease

13.2 Model Performance Tab

This section displays:

- Accuracy scores of all models
- Best-performing model
- Easy comparison view





14) Deployment

The final Streamlit application was deployed on **Render**.

This makes the project accessible online and demonstrates real-world deployment skills.

Deployment Benefits

- Easy sharing through resume/GitHub
- Recruiters can test the live project
- Demonstrates MLOps and production readiness

[Insert Deployment Screenshot]

App link- https://heart-disease-prediction-using-machine-ikso.onrender.com

15) Key Outcomes

- Successfully cleaned medically invalid values
- Performed strong EDA with business understanding
- Compared 4 ML classification models
- Achieved **86.41% accuracy** with Random Forest
- Built an end-to-end deployable ML application
- Published the project online

16) Conclusion

This project successfully demonstrates the complete lifecycle of a machine learning solution—from raw Kaggle data to a deployed web application.